

GENERATIVE ARTIFICIAL INTELLIGENCE FOR CHEMISTRY

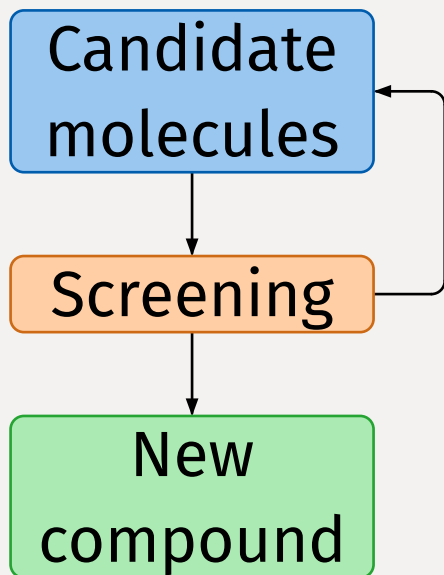
Hands-on session

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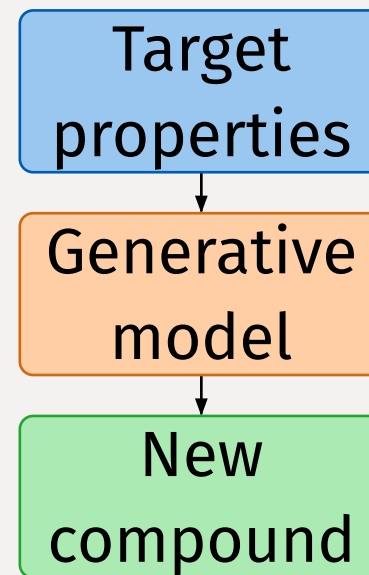


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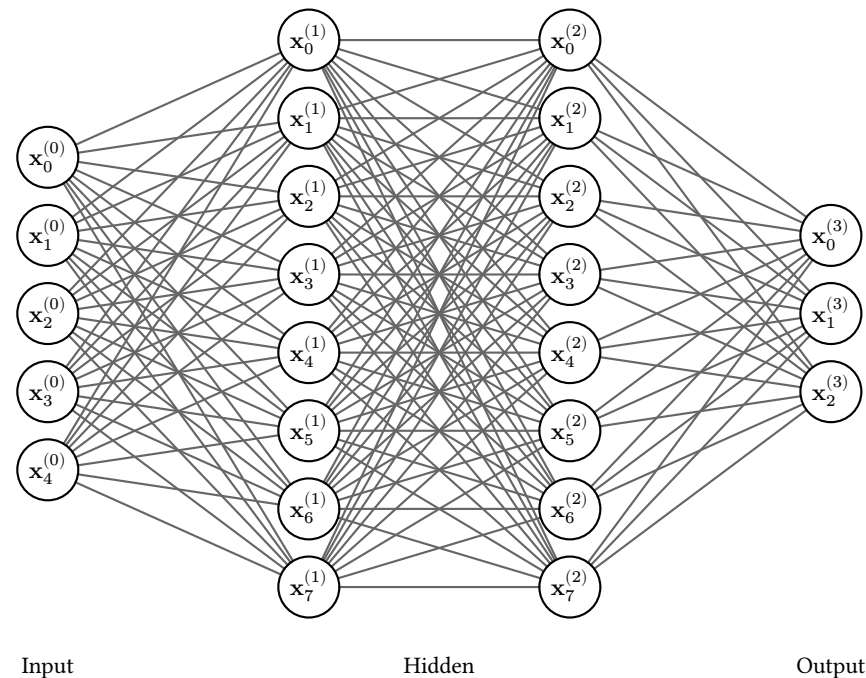
Forward design



Inverse design



- Allows models to learn complex intermediate representations
- Can fit arbitrary functions
- Gradient-based optimization
- For generative models:
 - What input and output?
 - How to define loss function?
/ What to optimize for?



$$\mathbf{x}^{(n)} = f(\mathbf{W}^{(n)}\mathbf{x}^{(n-1)} + \mathbf{b}^{(n)})$$

- Text-based, graph-structures or 3D coordinate based
- Commonly: **S**implified **M**olecular **I**nterpretation **L**ine **E**nter **S**ystem
 - Ethanol: CCO, OCC, C(C)O, ...
 - Phenol: c1ccccc1O, ...
 - Canonical (i.e. deterministic) SMILES possible
 - Augmentation: Include different SMILES for same molecule in training

- Define token vocabulary (e.g. c, n, 0, ...)
- Tokenization (e.g. CCO $\rightarrow$ [38, 38, 37])
- Input: Map each token to a vector
- Output: Probability for each token

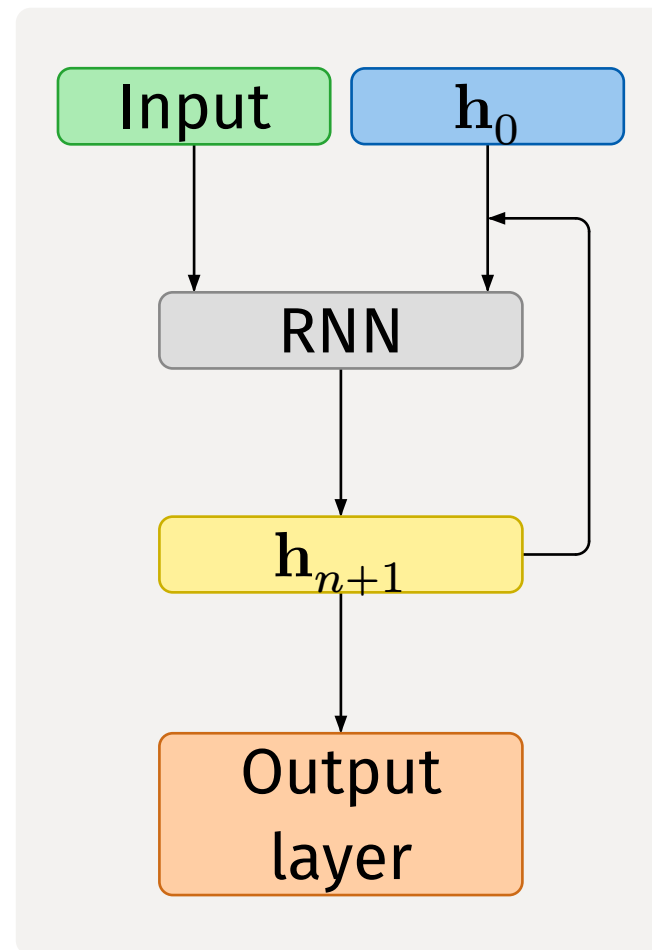
$$\text{softmax}(\mathbf{x}_i) = \frac{\exp(\mathbf{x}_i)}{\sum \exp(\mathbf{x}_j)}$$
$$\mathcal{L}_{\text{NLL}} = - \sum_{i=1}^N \log(\text{softmax}(\mathbf{x}_{i,c_i}))$$

Summarizing

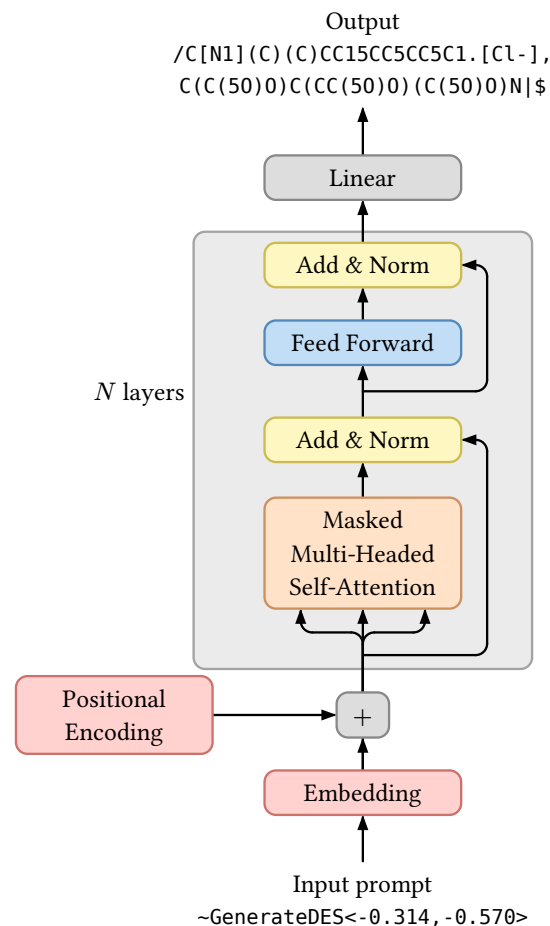
- Input tokens fed one at a time
 - Final hidden state serves as summary
 - Can be bidirectional
- Downstream regression / classification / ...

Generation

- First input is start-of-sequence token
- Predict next token probabilities based on h
- Newly decoded token is used as next input
 - In training: teacher-forcing
- Conditioning
 - Initial hidden state
 - Part of inputs

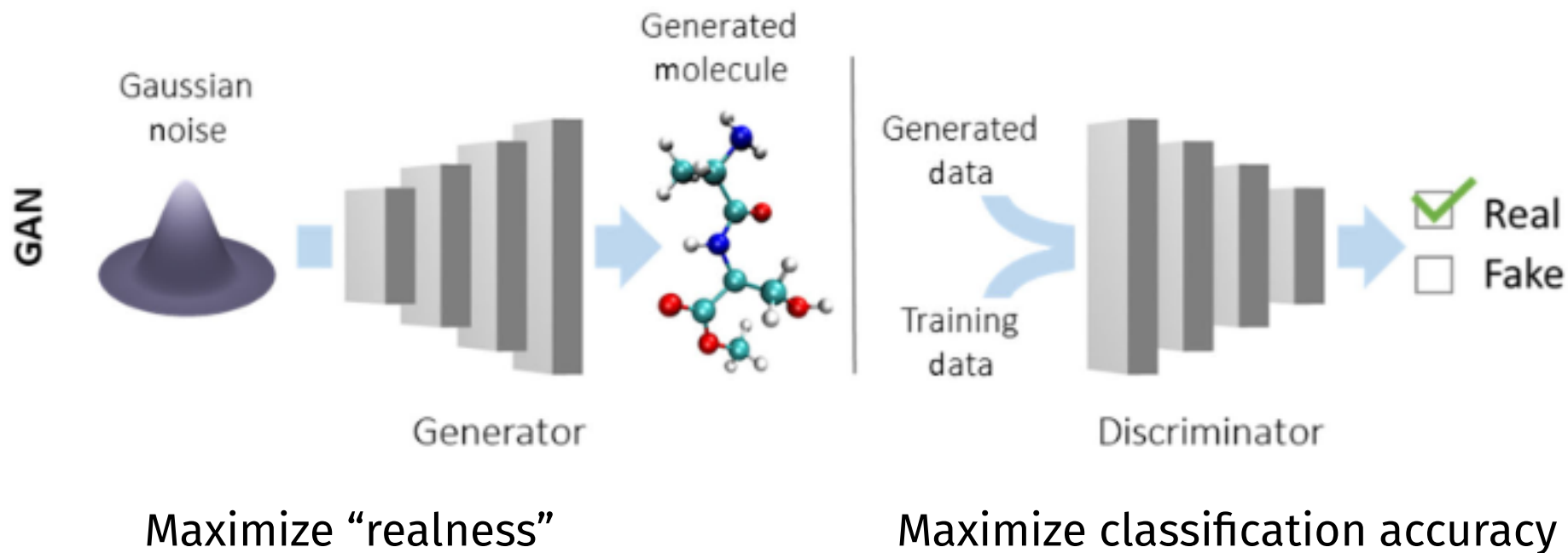


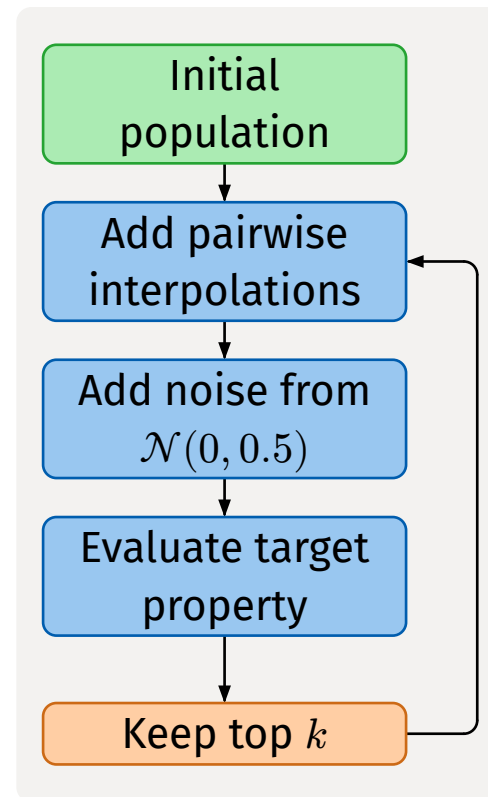
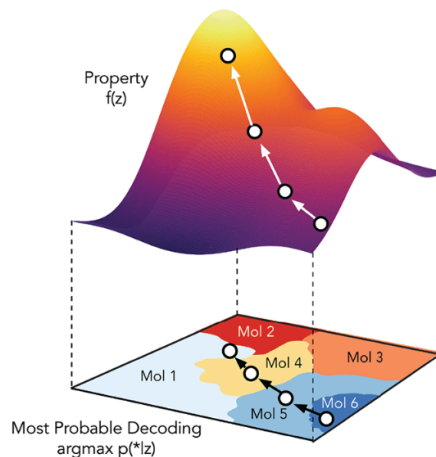
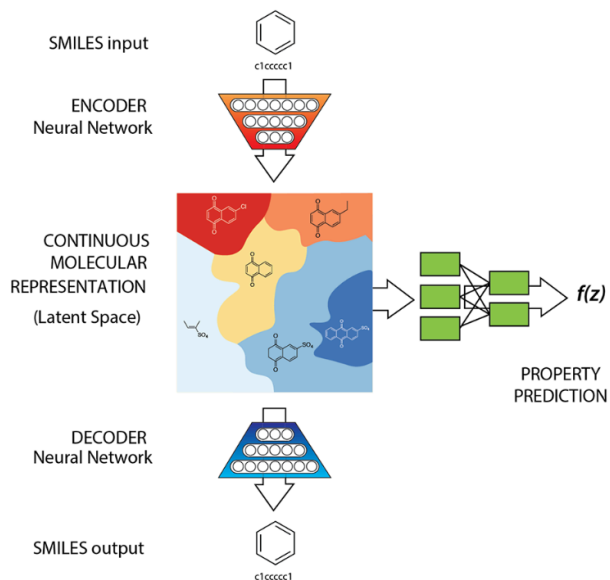
- Process sequences in parallel
- Attention mechanisms handle context
- Requires positional encoding



Luu, R. K.; Wysocki, M.; Buehler, M. J. Generative Discovery of de Novo Chemical Designs Using Diffusion Modeling and Transformer Deep Neural Networks with Application to Deep Eutectic Solvents. Appl. Phys. Lett. 2023, 122 (23), 234103. <https://doi.org/10.1063/5.0155890>.

- Generated SMILES may not be valid
 - Erroneous parentheses: CC(=OC, CC=O)C
 - Unclosed rings: c1cccc0
 - Impossible valence: CC(=O)(C)C
- Explicitly incorporate syntax / validity rules
- Self-referencing Embedded Strings (SELFIES)
 - Guaranteed validity





$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{pred}} + \beta D_{\text{KL}} + \dots$$

$$\mathbf{z} \sim \mathcal{N}(0, 1)$$

Gómez-Bombarelli, R.; Wei, J. N.; Duvenaud, D.; Hernández-Lobato, J. M.; Sánchez-Lengeling, B.; Sheberla, D.; Aguilera-Iparraguirre, J.; Hirzel, T. D.; Adams, R. P.; Aspuru-Guzik, A. Automatic Chemical Design Using a Data-Driven Continuous Representation of Molecules. ACS Cent. Sci. 2018, 4 (2), 268–276. <https://doi.org/10.1021/acscentsci.7b00572>.

- Diffusion
- Normalizing flow
- Evolutionary algorithms
- Reinforcement learning